Tehtävä 5

a) SELECT dept_name FROM departments;

```
mysql> SELECT dept_name FROM departments;
+--------------------+
| dept_name          |
+--------------------+
| Customer Service   |
| Development        |
| Finance            |
| Human Resources    |
| Marketing          |
| Production         |
| Quality Management |
| Research           |
| Sales              |
+--------------------+
9 rows in set (0.00 sec)
```

b) SELECT DISTINCT title FROM titles;

```
mysql> SELECT DISTINCT title FROM titles;
+--------------------+
| title              |
+--------------------+
| Senior Engineer    |
| Staff              |
| Engineer           |
| Senior Staff       |
| Assistant Engineer |
| Technique Leader   |
| Manager            |
+--------------------+
7 rows in set (0.15 sec)
```

c) SELECT MIN(salary), MAX(salary) FROM salaries;

```
mysql> SELECT MIN(salary), MAX(salary) FROM salaries;
+-------------+-------------+
| MIN(salary) | MAX(salary) |
+-------------+-------------+
|       38623 |      158220 |
+-------------+-------------+
1 row in set (0.76 sec)
```

d) SELECT AVG(salary) FROM salaries;

```
mysql> SELECT AVG(salary) FROM salaries;
+-------------+
| AVG(salary) |
+-------------+
|  63810.7448 |
+-------------+
1 row in set (1.01 sec)
```

e) SELECT * FROM employees WHERE(last_name)='Facello';

```
mysql> SELECT * FROM employees WHERE(last_name)='Facello';
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
|  10001 | 1953-09-02 | Georgi     | Facello   | M      | 1986-06-26 |
|  10327 | 1954-04-01 | Roded      | Facello   | M      | 1987-09-18 |
|  12751 | 1964-07-06 | Nahum      | Facello   | M      | 1995-01-09 |
|  15346 | 1959-09-26 | Kirk       | Facello   | F      | 1991-12-07 |
|  15685 | 1958-07-12 | Kasturi    | Facello   | M      | 1992-03-13 |
|  18686 | 1962-02-23 | Kwangyoen  | Facello   | F      | 1985-05-02 |
|  19041 | 1957-05-29 | Billur     | Facello   | F      | 1992-08-03 |
|  21947 | 1954-06-18 | Taisook    | Facello   | F      | 1991-07-30 |
|  23938 | 1955-07-11 | Nahum      | Facello   | M      | 1985-09-15 |
|  24774 | 1956-09-23 | Uno        | Facello   | F      | 1989-11-09 |
| 493549 | 1960-04-17 | Achilleas  | Facello   | F      | 1991-11-27 |
+--------+------------+------------+-----------+--------+------------+
186 rows in set (0.15 sec)
```

f) SELECT COUNT(*) FROM employees WHERE birth_date BETWEEN ‘1950-01-01’ AND ‘1959-12-31’;

```
mysql> SELECT COUNT(*) FROM employees WHERE birth_date BETWEEN '1950-01-01' AND '1959-12-31';
+----------+
| COUNT(*) |
+----------+
|   182886 |
+----------+
1 row in set (0.15 sec)

mysql>
```

g) SELECT gender, COUNT(*) FROM employees GROUP BY gender;

```
mysql> SELECT gender, COUNT(*) FROM employees GROUP BY gender;
+--------+----------+
| gender | COUNT(*) |
+--------+----------+
| M      |   179973 |
| F      |   120051 |
+--------+----------+
2 rows in set (0.17 sec)
```

Tehtävä 6

```
mysql> DESC pet;
+---------+-------------+------+-----+---------+----------------+
| Field   | Type        | Null | Key | Default | Extra          |
+---------+-------------+------+-----+---------+----------------+
| name    | varchar(20) | YES  |     | NULL    |                |
| species | varchar(20) | YES  |     | NULL    |                |
| sex     | char(1)     | YES  |     | NULL    |                |
| birth   | date        | YES  |     | NULL    |                |
| death   | date        | YES  |     | NULL    |                |
| petID   | int         | NO   | PRI | NULL    | auto_increment |
| ownerID | int         | YES  | MUL | NULL    |                |
+---------+-------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
mysql> DESC owner;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| first_name | varchar(255) | YES  |     | NULL    |                |
| last_name  | varchar(255) | YES  |     | NULL    |                |
| address    | varchar(255) | YES  |     | NULL    |                |
| ownerID    | int          | NO   | PRI | NULL    | auto_increment |
+------------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

Tarvittavat MySQL lauseet:

```
CREATE TABLE owner
        -> (
        -> first_name varchar(255),
        -> last_name varchar(255),
        -> address varchar(255),
        -> ownerID int PRIMARY KEY AUTO_INCREMENT
        -> ) ;
ALTER TABLE pet DROP COLUMN owner;
ALTER TABLE pet ADD COLUMN petID INT PRIMARY KEY AUTO_INCREMENT;
ALTER TABLE pet ADD COLUMN ownerID INT;
ALTER TABLE pet ADD CONSTRAINT FOREIGN KEY(ownerID) REFERENCES owner(ownerID);
INSERT INTO owner VALUES ('Harold', 'Lloyd', 'Someaddress 1', NULL);
UPDATE pet SET ownerID=1 WHERE name='Fluffy' OR name='Buffy';
DESC owner;
DESC pet;
```

Tehtävä 7

a) SELECT * FROM employees ORDER BY last_name LIMIT 10;

```
mysql> SELECT * FROM employees ORDER BY last_name LIMIT 10;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
|  11761 | 1964-07-17 | Bartek     | Aamodt    | M      | 1991-06-12 |
|  15427 | 1959-03-06 | Aluzio     | Aamodt    | M      | 1985-03-03 |
|  18182 | 1963-02-23 | Dekang     | Aamodt    | F      | 1988-05-25 |
|  16572 | 1956-11-26 | Matt       | Aamodt    | M      | 1987-06-16 |
|  12791 | 1960-06-16 | Mokhtar    | Aamodt    | M      | 1994-08-14 |
|  12516 | 1958-06-25 | Sreenivas  | Aamodt    | F      | 1990-03-06 |
|  12982 | 1952-12-08 | Sachem     | Aamodt    | F      | 1992-01-11 |
|  17400 | 1962-03-22 | Basim      | Aamodt    | F      | 1991-09-15 |
|  19898 | 1957-03-09 | Vidar      | Aamodt    | M      | 1988-08-06 |
|  17885 | 1954-02-01 | Takanari   | Aamodt    | M      | 1996-08-19 |
+--------+------------+------------+-----------+--------+------------+
10 rows in set (0.19 sec)
```

b) SELECT * FROM employees ORDER BY last_name, first_name LIMIT 10;

```
mysql> SELECT * FROM employees ORDER BY last_name, first_name LIMIT 10;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
| 258641 | 1961-05-23 | Abdelkader | Aamodt    | M      | 1994-12-02 |
| 258005 | 1953-02-17 | Adhemar    | Aamodt    | F      | 1991-01-21 |
| 455773 | 1960-05-04 | Aemilian   | Aamodt    | M      | 1988-04-21 |
| 436560 | 1959-03-16 | Alagu      | Aamodt    | F      | 1991-10-17 |
| 266651 | 1959-05-28 | Aleksander | Aamodt    | F      | 1989-03-29 |
| 487598 | 1962-03-03 | Alexius    | Aamodt    | M      | 1994-12-30 |
| 216963 | 1960-07-16 | Alois      | Aamodt    | M      | 1995-08-24 |
|  15427 | 1959-03-06 | Aluzio     | Aamodt    | M      | 1985-03-03 |
| 100860 | 1964-06-20 | Amabile    | Aamodt    | F      | 1993-02-06 |
| 107070 | 1954-04-24 | Anestis    | Aamodt    | M      | 1990-10-30 |
+--------+------------+------------+-----------+--------+------------+
10 rows in set (0.22 sec)
```

c) ELECT * FROM employees ORDER BY hire_date DESC, first_name LIMIT 5;

```
mysql> SELECT * FROM employees ORDER BY hire_date DESC, first_name LIMIT 5;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
| 463807 | 1964-06-12 | Bikash     | Covnot    | M      | 2000-01-28 |
| 428377 | 1957-05-09 | Yucai      | Gerlach   | M      | 2000-01-23 |
| 499553 | 1954-05-06 | Hideyuki   | Delgrande | F      | 2000-01-22 |
| 222965 | 1959-08-07 | Volkmar    | Perko     | F      | 2000-01-13 |
|  47291 | 1960-09-09 | Ulf        | Flexer    | M      | 2000-01-12 |
+--------+------------+------------+-----------+--------+------------+
5 rows in set (0.15 sec)
```

d) SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no ORDER BY salary DESC LIMIT 1;

```
mysql> SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no ORDER BY
salary DESC LIMIT 1;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Tokuyasu   | Pesch     | 158220 |
+------------+-----------+--------+
1 row in set (2.65 sec)
```

e) SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no ORDER BY salary ASC LIMIT 1;

```
mysql> SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no ORDER BY
salary ASC LIMIT 1;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Olivera    | Baek      |  38623 |
+------------+-----------+--------+
1 row in set (2.57 sec)
```

f) SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no WHERE salary>150000 ORDER BY salary;

```
mysql> SELECT first_name, last_name, salary FROM employees JOIN salaries ON employees.emp_no = salaries.emp_no WHERE salary>150000 ORDER BY salary;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Lansing    | Kambil    | 150052 |
| Ibibia     | Junet     | 150345 |
| Honesty    | Mukaidono | 156286 |
| Tokuyasu   | Pesch     | 157821 |
| Tokuyasu   | Pesch     | 158220 |
+------------+-----------+--------+
36 rows in set (1.02 sec)
```

g) SELECT COUNT(*) AS Sales FROM employees
-> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
-> JOIN departments ON dept_emp.dept_no = departments.dept_no
-> Where dept_name = 'Sales';

```
mysql> SELECT COUNT(*) AS Sales FROM employees
    -> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
    -> JOIN departments ON dept_emp.dept_no = departments.dept_no
    -> Where dept_name = 'Sales';
+-------+
| Sales |
+-------+
| 52245 |
+-------+
1 row in set (0.21 sec)
```

SELECT COUNT(*) AS Marketing FROM employees
-> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
-> JOIN departments ON dept_emp.dept_no = departments.dept_no
-> Where dept_name = 'Marketing';

```
mysql> SELECT COUNT(*) AS Marketing FROM employees
    -> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
    -> JOIN departments ON dept_emp.dept_no = departments.dept_no
    -> Where dept_name = 'Marketing';
+-----------+
| Marketing |
+-----------+
|     20211 |
+-----------+
1 row in set (0.14 sec)
```

h) SELECT first_name, last_name, dept_name
-> FROM employees
-> JOIN dept_manager ON employees.emp_no = dept_manager.emp_no
-> JOIN departments ON dept_manager.dept_no = departments.dept_no;

```
mysql> SELECT first_name, last_name, dept_name
    -> FROM employees
    -> JOIN dept_manager ON employees.emp_no = dept_manager.emp_no
    -> JOIN departments ON dept_manager.dept_no = departments.dept_no;
+------------+--------------+--------------------+
| first_name | last_name    | dept_name          |
+------------+--------------+--------------------+
| Tonny      | Butterworth  | Customer Service   |
| Marjo      | Giarratana   | Customer Service   |
| Xiaobin    | Spinelli     | Customer Service   |
| Yuchang    | Weedman      | Customer Service   |
| DeForest   | Hagimont     | Development        |
| Leon       | DasSarma     | Development        |
| Ebru       | Alpin        | Finance            |
| Isamu      | Legleitner   | Finance            |
| Shirish    | Ossenbruggen | Human Resources    |
| Karsten    | Sigstam      | Human Resources    |
| Margareta  | Markovitch   | Marketing          |
| Vishwani   | Minakawa     | Marketing          |
| Krassimir  | Wegerle      | Production         |
| Rosine     | Cools        | Production         |
| Shem       | Kieras       | Production         |
| Oscar      | Ghazalie     | Production         |
| Peternela  | Onuegbe      | Quality Management |
| Rutger     | Hofmeyr      | Quality Management |
| Sanjoy     | Quadeer      | Quality Management |
| Dung       | Pesch        | Quality Management |
| Arie       | Staelin      | Research           |
| Hilary     | Kambil       | Research           |
| Przemyslawa | Kaelbling   | Sales              |
| Hauke      | Zhang        | Sales              |
+------------+--------------+--------------------+
24 rows in set (0.02 sec)
```

i) SELECT AVG(salary) AS Sales_AVG_Salary
-> FROM employees
-> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
-> JOIN departments ON dept_emp.dept_no = departments.dept_no
-> JOIN salaries ON employees.emp_no = salaries.emp_no
-> WHERE dept_name = 'Sales';

```
mysql> SELECT AVG(salary) AS Sales_AVG_Salary
    -> FROM employees
    -> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
    -> JOIN departments ON dept_emp.dept_no = departments.dept_no
    -> JOIN salaries ON employees.emp_no = salaries.emp_no
    -> WHERE dept_name = 'Sales';
+------------------+
| Sales_AVG_Salary |
+------------------+
|       80667.6058 |
+------------------+
1 row in set (0.89 sec)
```

SELECT AVG(salary) AS Marketing_AVG_Salary
-> FROM employees
-> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
-> JOIN departments ON dept_emp.dept_no = departments.dept_no
-> JOIN salaries ON employees.emp_no = salaries.emp_no
-> WHERE dept_name = 'Marketing';

```
mysql> SELECT AVG(salary) AS Marketing_AVG_Salary
    -> FROM employees
    -> JOIN dept_emp ON employees.emp_no = dept_emp.emp_no
    -> JOIN departments ON dept_emp.dept_no = departments.dept_no
    -> JOIN salaries ON employees.emp_no = salaries.emp_no
    -> WHERE dept_name = 'Marketing';
+----------------------+
| Marketing_AVG_Salary |
+----------------------+
|           71913.2000 |
+----------------------+
1 row in set (0.49 sec)
```

Tehtävä 8

a) SELECT name FROM language ORDER BY name;

```
mysql> SELECT name FROM language ORDER BY name;
+-----------+
| name      |
+-----------+
| English   |
| French    |
| German    |
| Italian   |
| Japanese  |
| Mandarin  |
+-----------+
6 rows in set (0.03 sec)
```

b) SELECT last_name, title
-> FROM actor
-> JOIN film_actor on actor.actor_id = film_actor.actor_id
-> JOIN film ON film_actor.film_id = film.film_id
-> WHERE last_name = 'Temple';

```
mysql> SELECT last_name, title
    -> FROM actor
    -> JOIN film_actor on actor.actor_id = film_actor.actor_id
    -> JOIN film ON film_actor.film_id = film.film_id
    -> WHERE last_name = 'Temple';
+-----------+-------------------------+
| last_name | title                   |
+-----------+-------------------------+
| TEMPLE    | ACADEMY DINOSAUR        |
| TEMPLE    | ALABAMA DEVIL           |
| TEMPLE    | BALLOON HOMEWARD        |
| TEMPLE    | VIRGINIAN PLUTO         |
| TEMPLE    | WARDROBE PHANTOM        |
| TEMPLE    | WRONG BEHAVIOR          |
+-----------+-------------------------+
104 rows in set (0.00 sec)
```

c) SELECT first_name, last_name
-> FROM film
-> JOIN film_actor ON film.film_id = film_actor.film_id
-> JOIN actor ON film_actor.actor_id = actor.actor_id
-> WHERE title = 'Ghost Groundhog';

```
mysql> SELECT first_name, last_name
    -> FROM film
    -> JOIN film_actor ON film.film_id = film_actor.film_id
    -> JOIN actor ON film_actor.actor_id = actor.actor_id
    -> WHERE title = 'Ghost Groundhog';
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| DAN        | HARRIS    |
| KENNETH    | TORN      |
| KEVIN      | GARLAND   |
| RUSSELL    | TEMPLE    |
| RENEE      | BALL      |
+------------+-----------+
5 rows in set (0.01 sec)
```

d) SELECT COUNT(*) AS Horror_Movies
-> FROM category
-> JOIN film_category ON category.category_id = film_category.category_id
-> WHERE name = 'Horror';

```
mysql> SELECT COUNT(*) AS Horror_Movies
    -> FROM category
    -> JOIN film_category ON category.category_id = film_category.category_id
    -> WHERE name = 'Horror';
+----------------+
| Horror_Movies |
+----------------+
|            56 |
+----------------+
1 row in set (0.02 sec)
```

e) SELECT title
-> FROM category
-> JOIN film_category ON category.category_id = film_category.category_id
-> JOIN film ON film_category.film_id = film.film_id
-> WHERE name = 'Horror';

```
mysql> SELECT title
    -> FROM category
    -> JOIN film_category ON category.category_id = film_category.category_id
    -> JOIN film ON film_category.film_id = film.film_id
    -> WHERE name = 'Horror';
+----------------------------+
| title                      |
+----------------------------+
| ACE GOLDFINGER             |
| AFFAIR PREJUDICE           |
| AIRPORT POLLOCK            |
| ALABAMA DEVIL              |
| ALI FOREVER                |
| ANALYZE HOOSIERS           |
| ANYTHING SAVANNAH          |
| ARABIA DOGMA               |
| ARACHNOPHOBIA ROLLERCOASTER |
| BEHAVIOR RUNAWAY           |
| BOWFINGER GABLES           |
| CARRIE BUNCH               |
| COMMANDMENTS EXPRESS       |
| DESERT POSEIDON            |
| DRUMS DYNAMITE             |
| EGYPT TENENBAUMS           |
| ELEPHANT TROJAN            |
| FAMILY SWEET               |
| FIDELITY DEVIL             |
| FREDDY STORM               |
| GASLIGHT CRUSADE           |
| HIGH ENCINO                |
| JAPANESE RUN               |
| KARATE MOON                |
| KENTUCKIAN GIANT           |
| LADY STAGE                 |
| LOLA AGENT                 |
| LOVE SUICIDES              |
| MONTEREY LABYRINTH         |
| MOTIONS DETAILS            |
| PANIC CLUB                 |
| PARIS WEEKEND              |
| PATTON INTERVIEW           |
| PULP BEVERLY               |
| REAP UNFAITHFUL            |
| REEF SALUTE                |
| ROCK INSTINCT              |
| ROLLERCOASTER BRINGING     |
| RULES HUMAN                |
| SIMON NORTH                |
| SINNERS ATLANTIS           |
| SLEEPING SUSPECTS          |
| SPIRIT FLINTSTONES         |
| STRANGERS GRAFFITI         |
| STREETCAR INTENTIONS       |
| SWARM GOLD                 |
| TARZAN VIDEOTAPE           |
| TEMPLE ATTRACTION          |
| TEXAS WATCH                |
| TRAIN BUNCH                |
| TREASURE COMMAND           |
| UNDEFEATED DALMATIONS      |
| WATERSHIP FRONTIER         |
| WORLD LEATHERNECKS         |
| YENTL IDAHO                |
| ZHIVAGO CORE               |
+----------------------------+
56 rows in set (0.00 sec)
```